

Demographic Forecasting

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Harvard University

Joint work with Federico Girosi (RAND)
with contributions from Kevin Quinn and Gregory Wawro

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- Approach: Formalizing **qualitative** insights in **quantitative** models

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- Better ways of creating Bayesian priors
- Produces forecasts and farcasts using considerably more information

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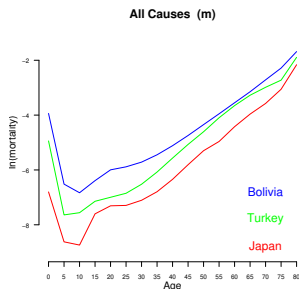
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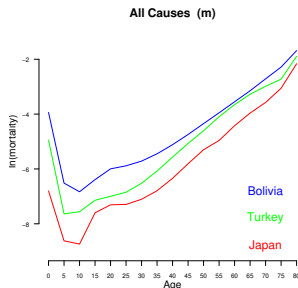
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Existing Method 1: Parameterize the Age Profile

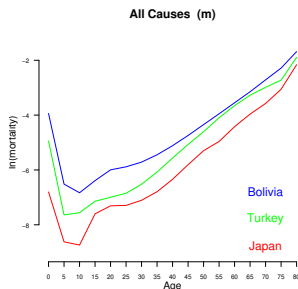


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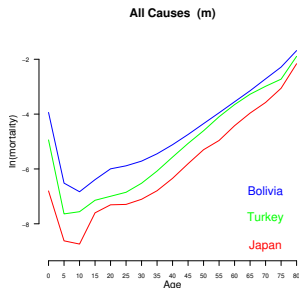
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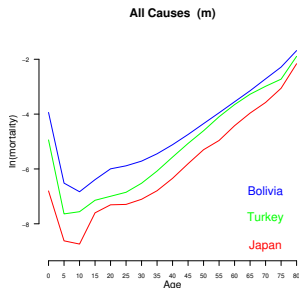
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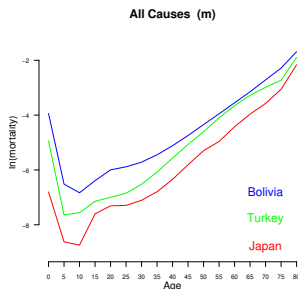
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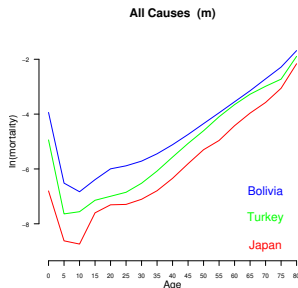
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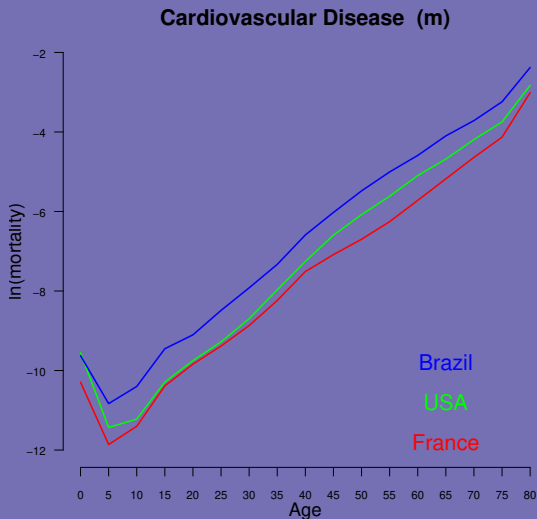
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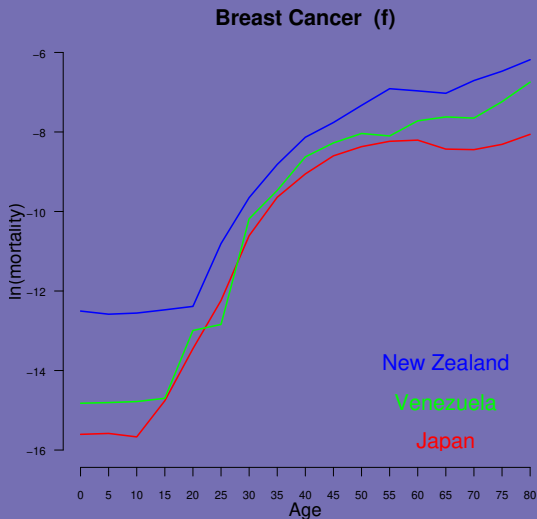


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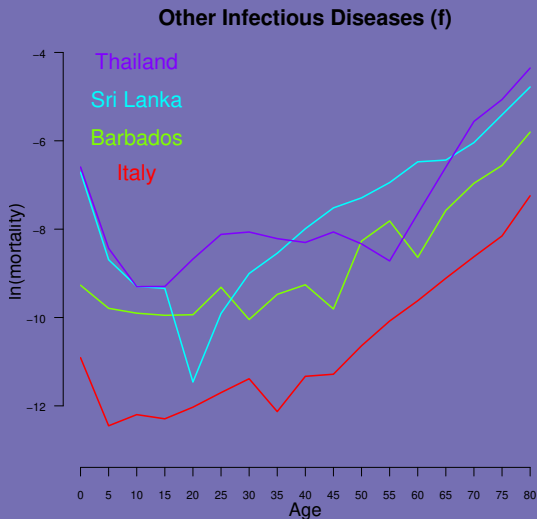
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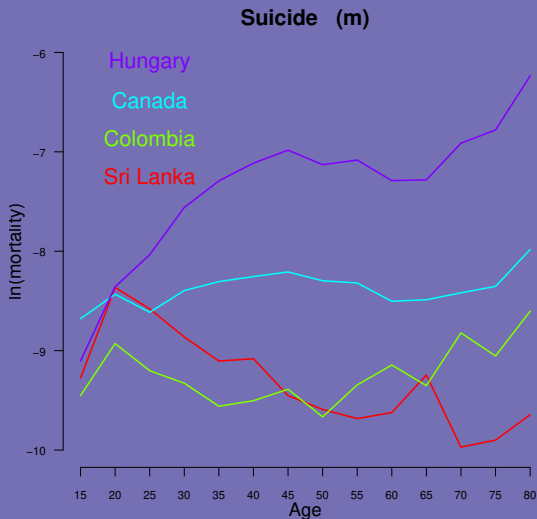
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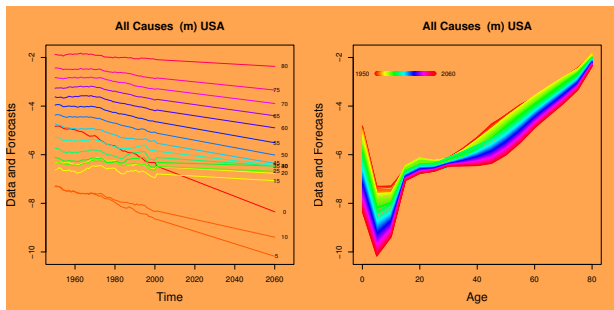
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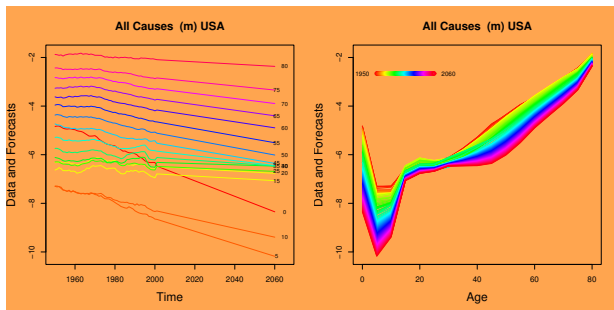
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- Also: Method ignores covariate information; the leading current method (McNown-Rogers) not replicable

Existing Method 2: Deterministic Projections

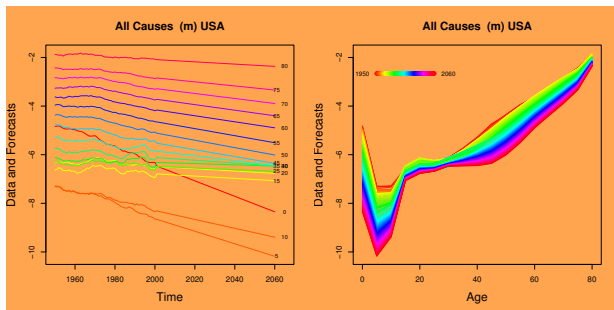


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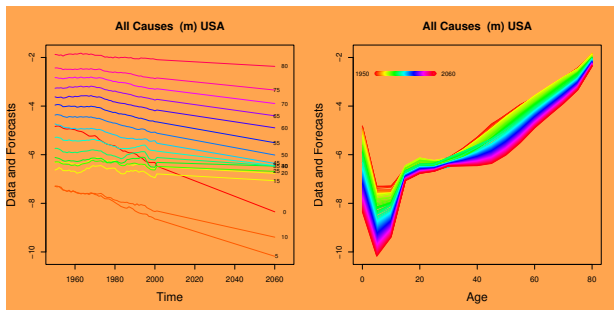
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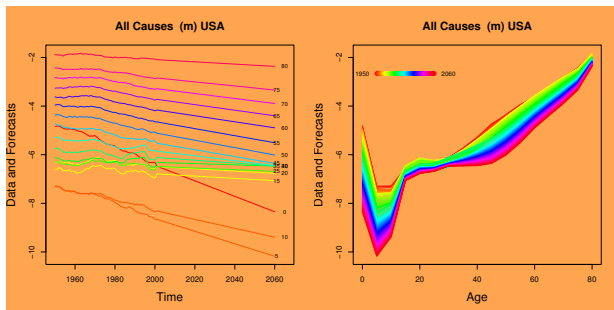
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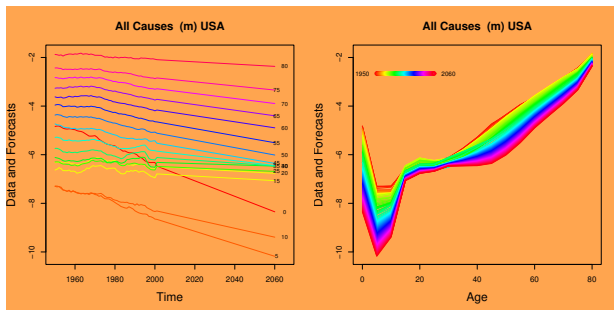
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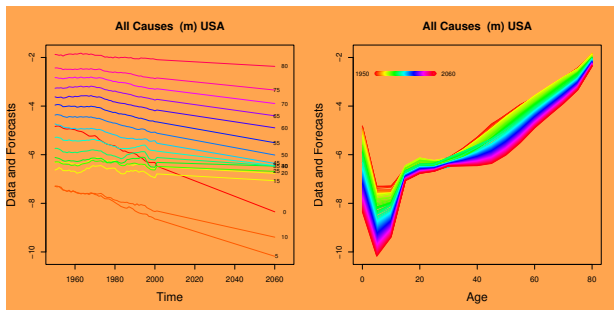
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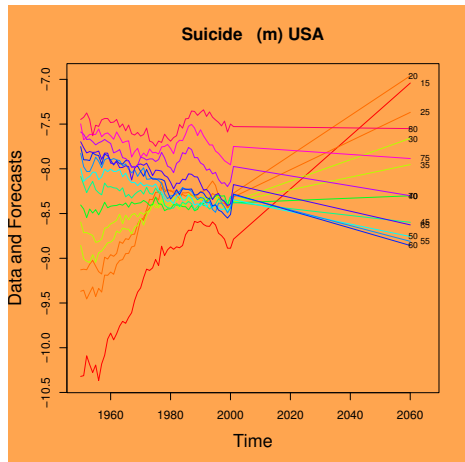
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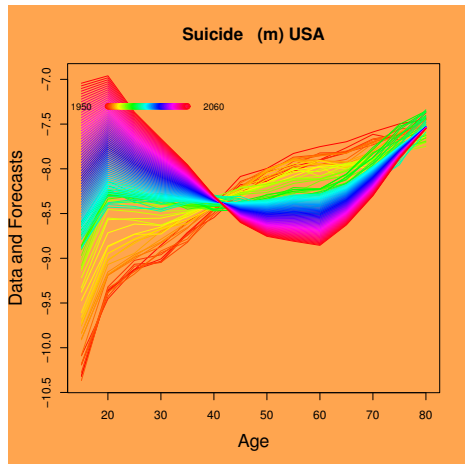
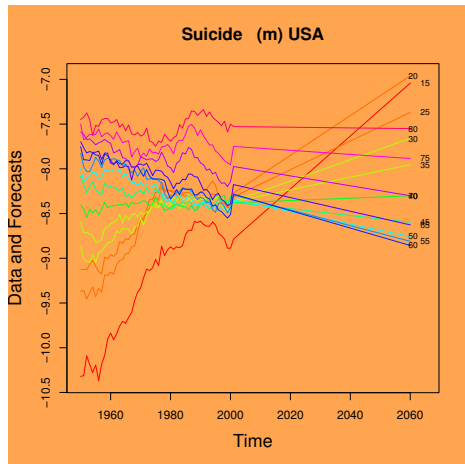
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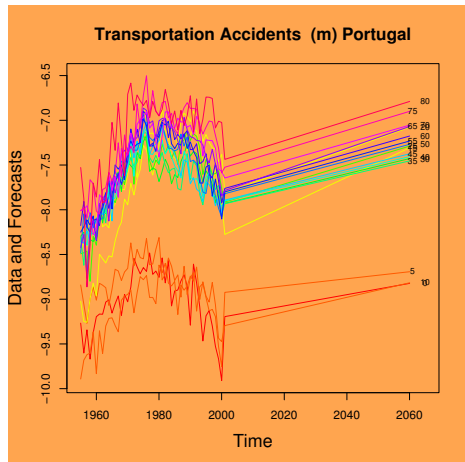
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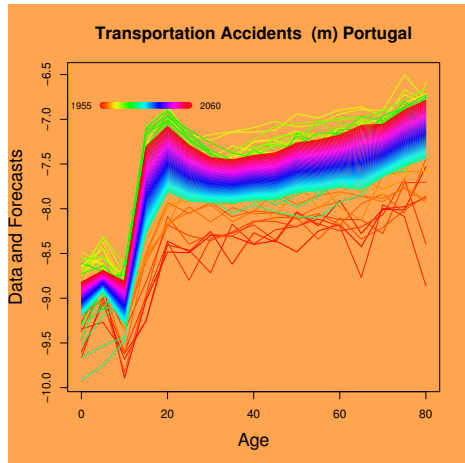
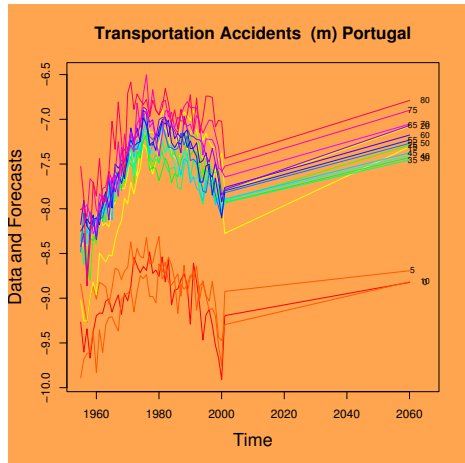
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- Likelihood for equation-by-equation least squares:

$$\mathcal{P}(m \mid \beta_i, \sigma_i) = \prod_t \mathcal{N}(m_{it} \mid \mathbf{z}_{it}\beta_i, \sigma_i^2)$$

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- The easy part: *easy-to-use software* to implement everything we discuss today.

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Natural choice for the prior:

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- Extensive trial-and-error runs, yielded no useful parameter values.

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 - 3 In the subspace, we can invert $\mu = \mathbf{Z}\beta$ as $\beta = (\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mu$, giving:

$$\mathcal{P}(\beta \mid \theta) \propto \exp\left(-\frac{1}{2}H[\mu, \theta]\right) = \exp\left(-\frac{1}{2}H[\mathbf{Z}\beta, \theta]\right)$$

the same prior on μ , expressed as a function of β (with constant Jacobian).

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- We start with one-dimensional $P(\mu_{cat})$, and treat it as the multidimensional $P(\beta_{ca})$, constant in all directions but $Z_{cat}\beta_{ca}$.

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- Priors are based on knowledge rather than guesses.

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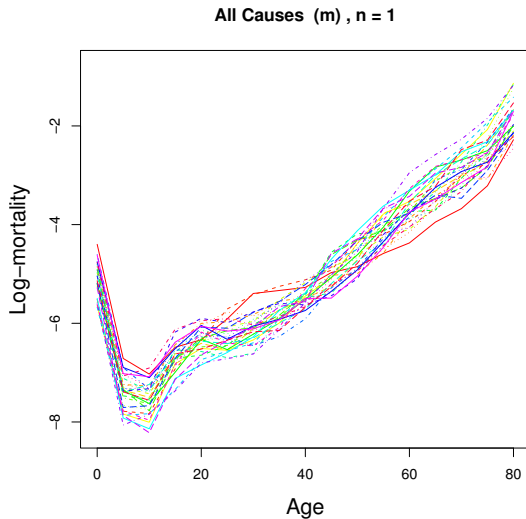
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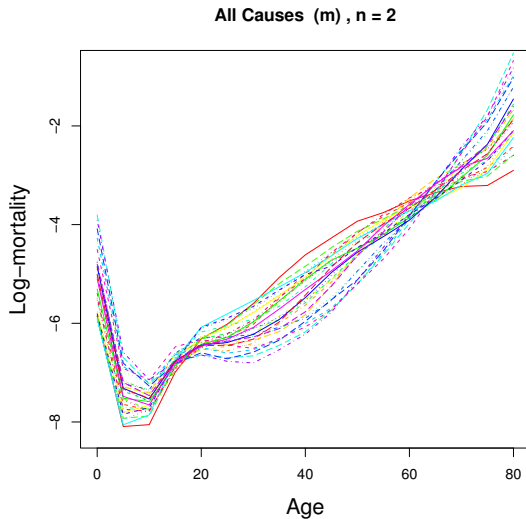
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- Different age groups can have different covariates: the matrices $\mathbf{C}_{aa'} \equiv \frac{1}{T} \mathbf{Z}'_a \mathbf{Z}_{a'}$ are rectangular ($d_a \times d_{a'}$).

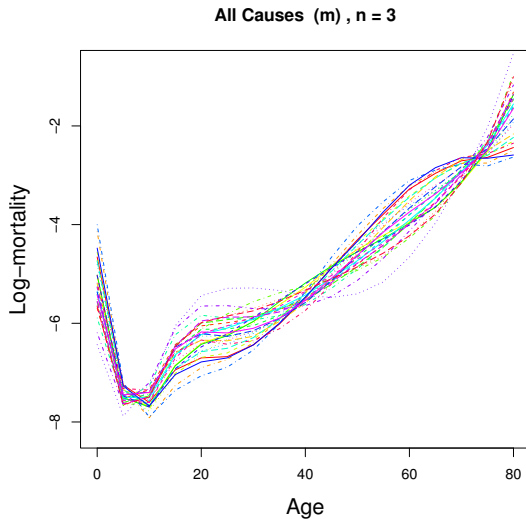
Samples From Age Prior



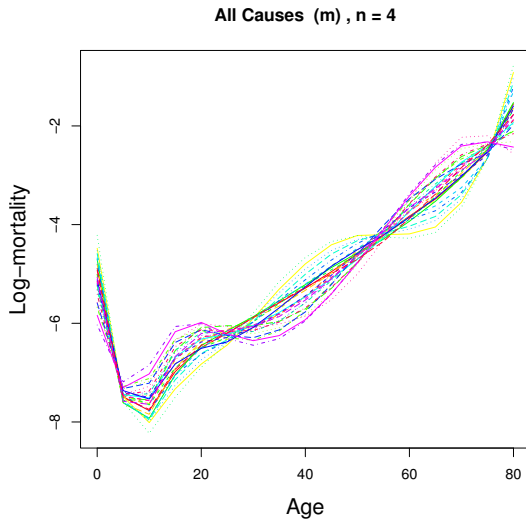
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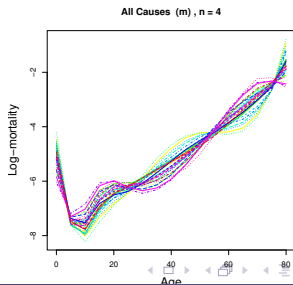
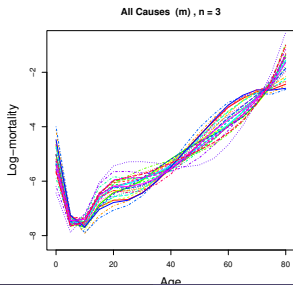
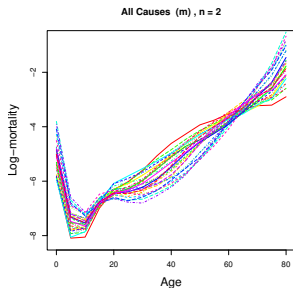
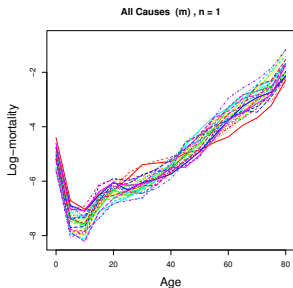
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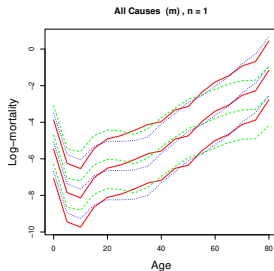
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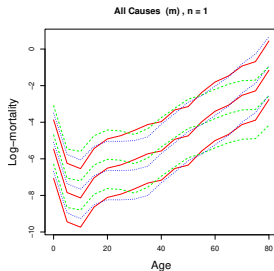
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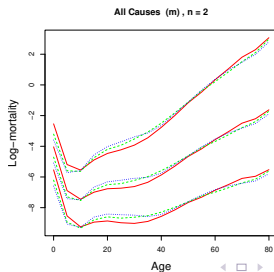
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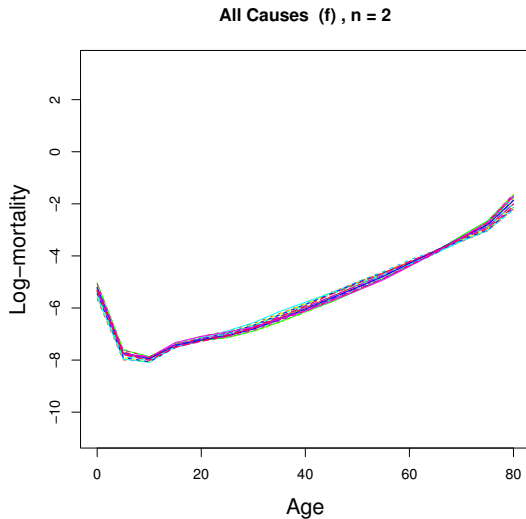
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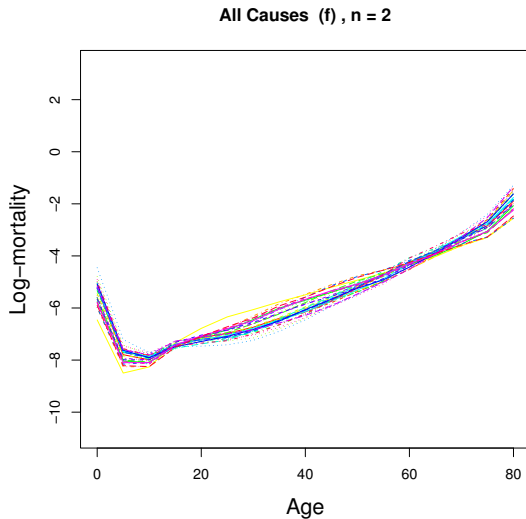
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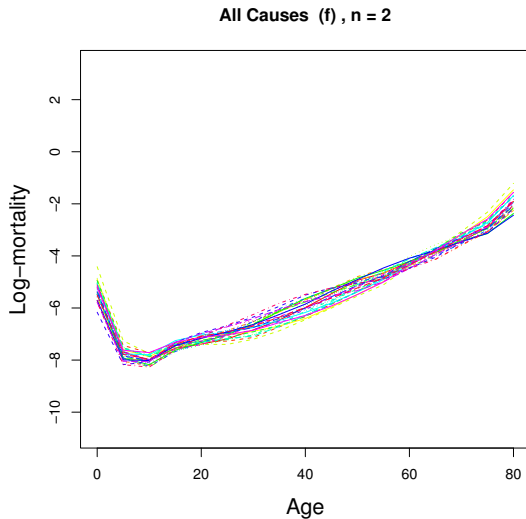
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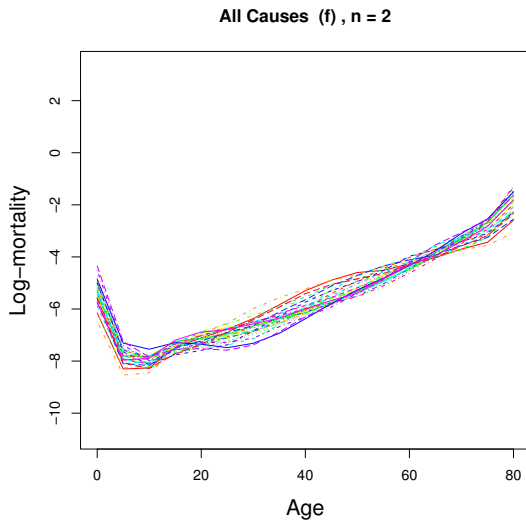
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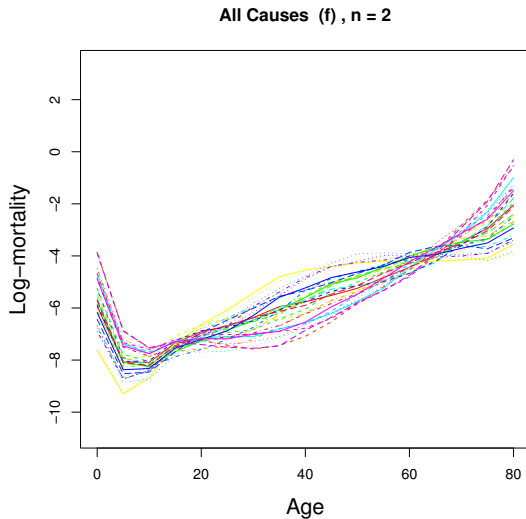
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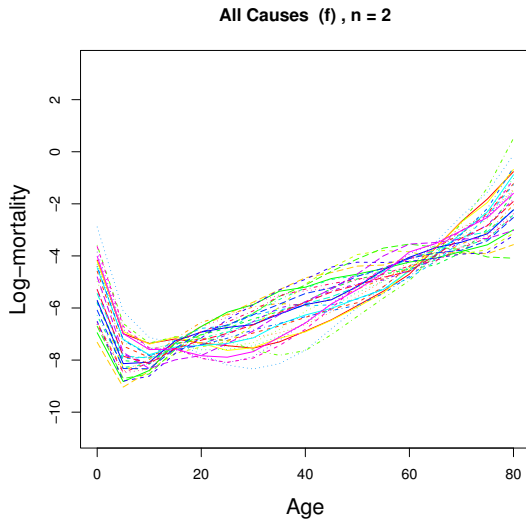
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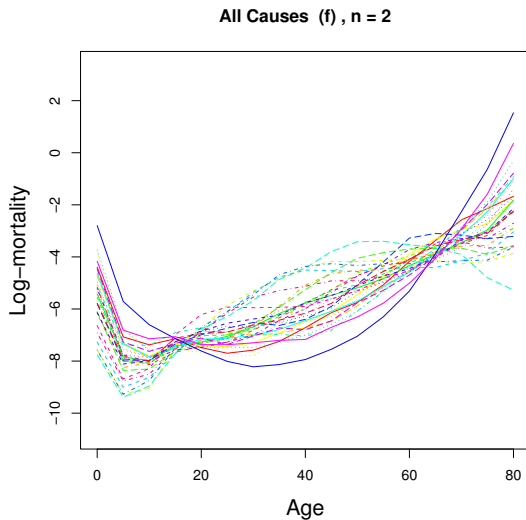
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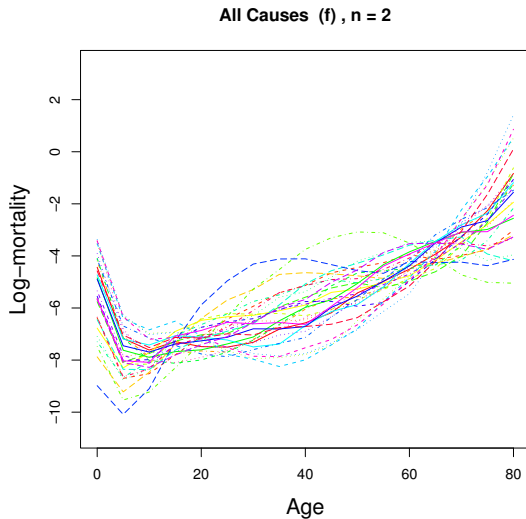
Samples from Age Prior



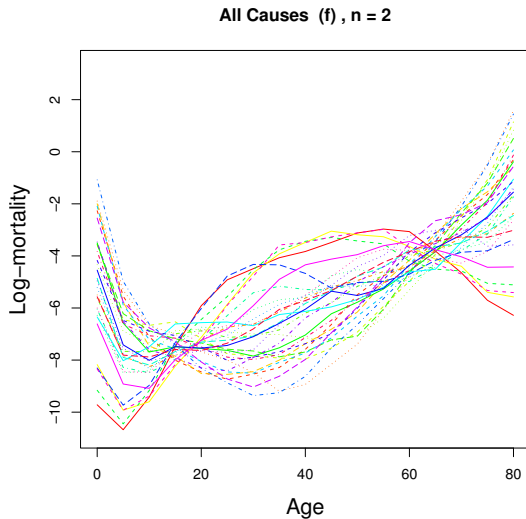
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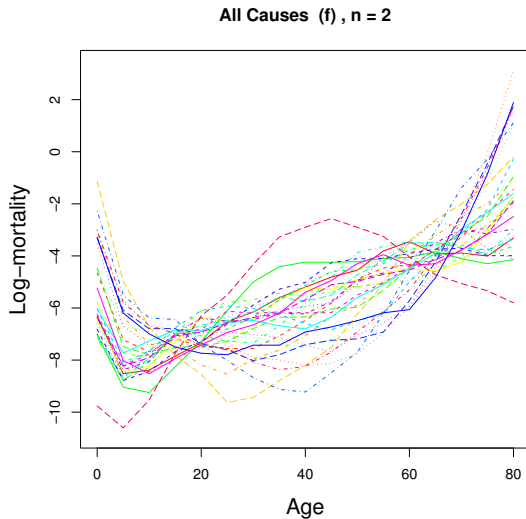
Samples from Age Prior



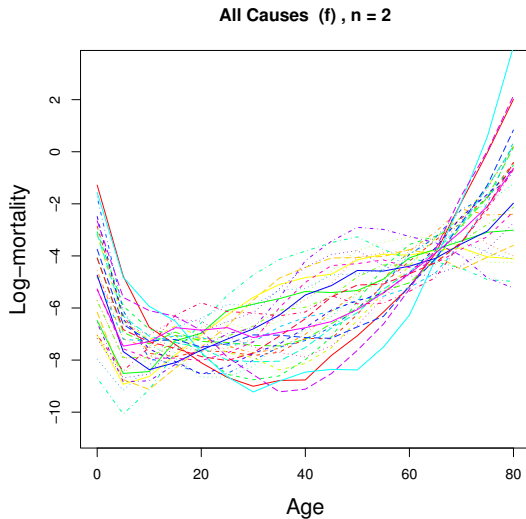
Samples from Age Prior



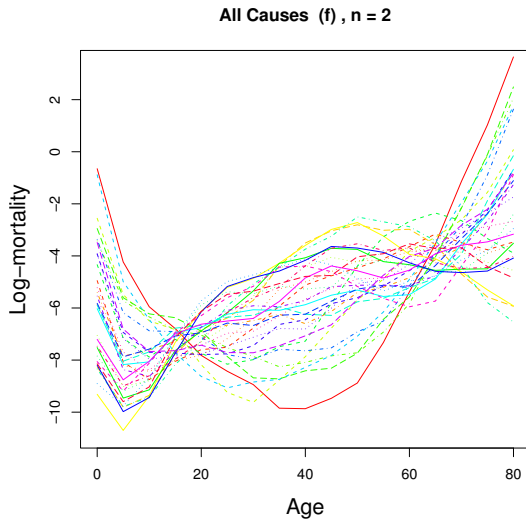
Samples from Age Prior



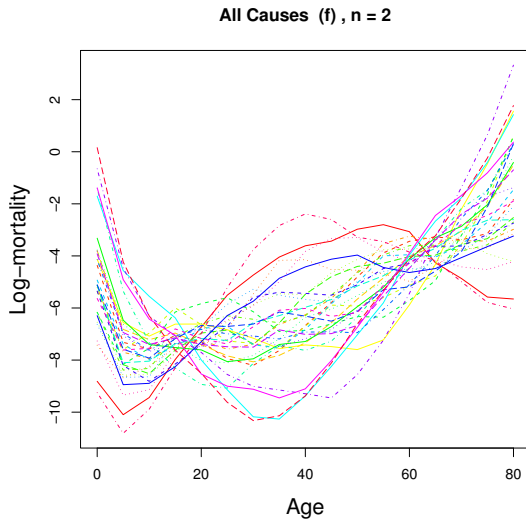
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Generalizations

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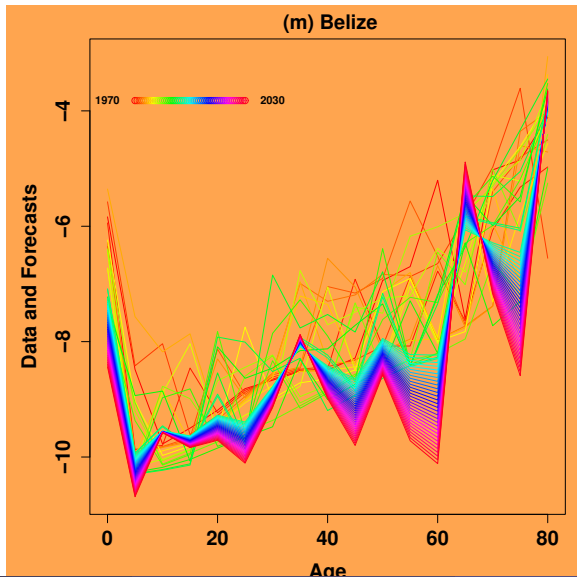
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- The mathematical form for *all* these (separately or together) turns out to be the same:

$$\mathcal{P}(\boldsymbol{\beta} \mid \theta) \propto \exp \left(-\frac{\theta}{2} \sum_{ij} W_{ij} \boldsymbol{\beta}_i' \mathbf{C}_{ij} \boldsymbol{\beta}_j \right), \quad \mathbf{C}_{aa'} \equiv \frac{1}{T} \mathbf{Z}_a \mathbf{Z}_{a'}$$

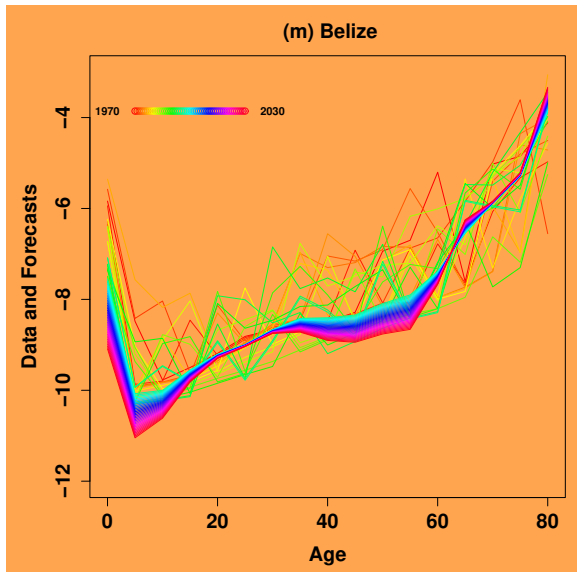
Mortality from Respiratory Infections, Males

Least Squares



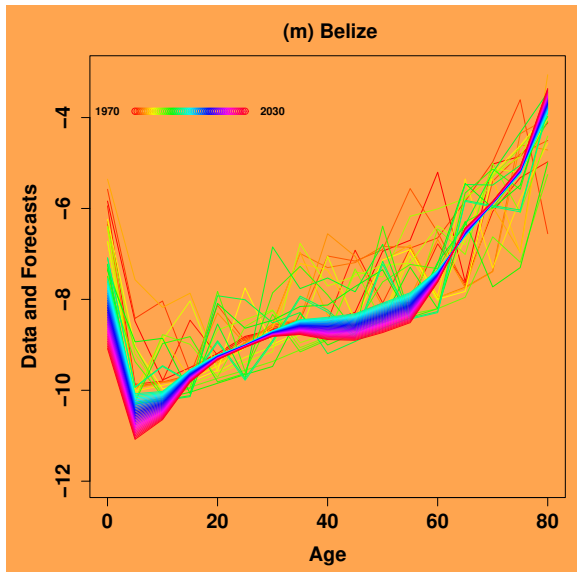
Mortality from Respiratory Infections, males, $\sigma = 2.00$

Smoothing over Age Groups



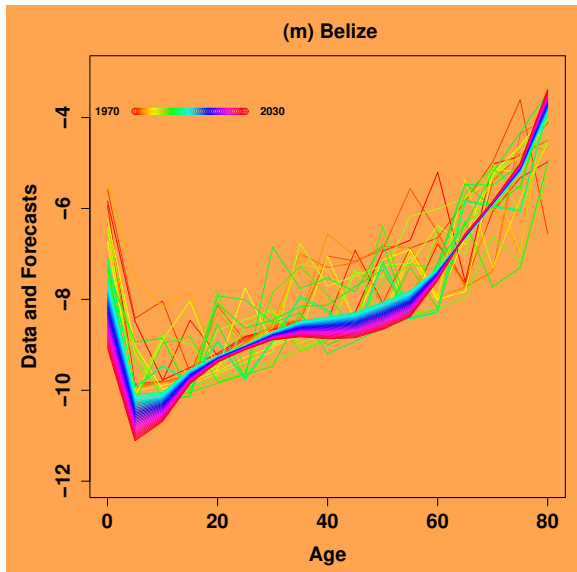
Mortality from Respiratory Infections, males, $\sigma = 1.51$

Smoothing over Age Groups



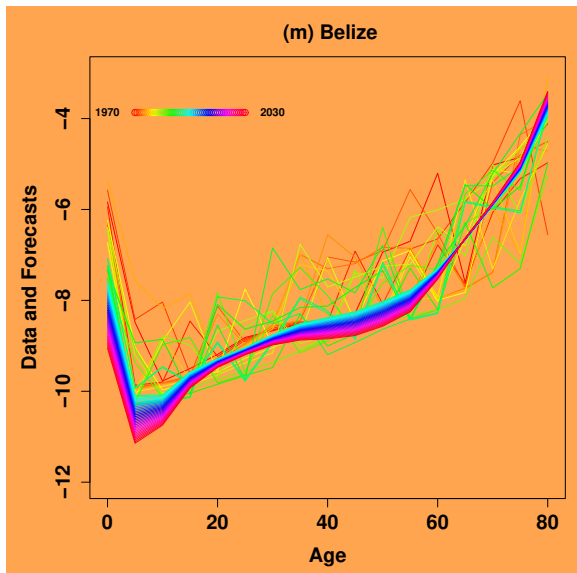
Mortality from Respiratory Infections, males, $\sigma = 1.15$

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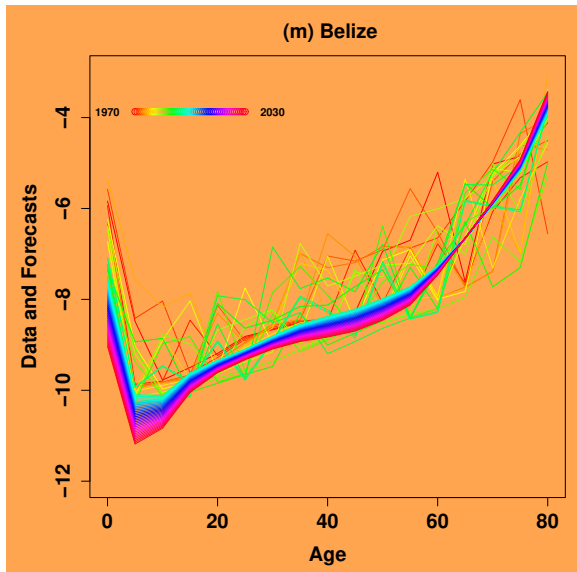
Mortality from Respiratory Infections, males, $\sigma = 0.87$

Smoothing over Age Groups



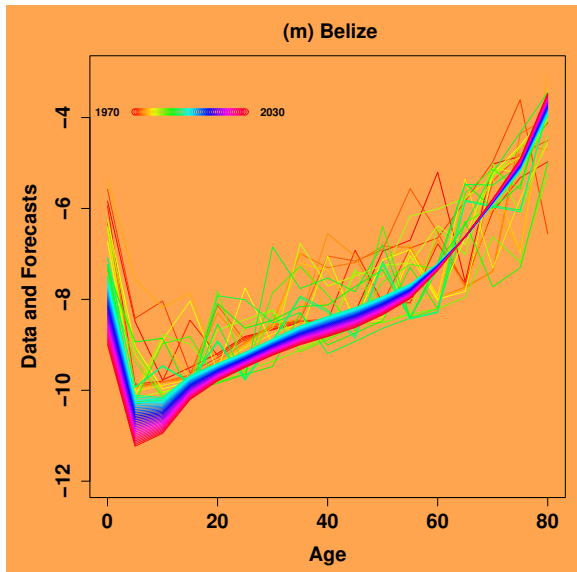
Mortality from Respiratory Infections, males, $\sigma = 0.66$

Smoothing over Age Groups



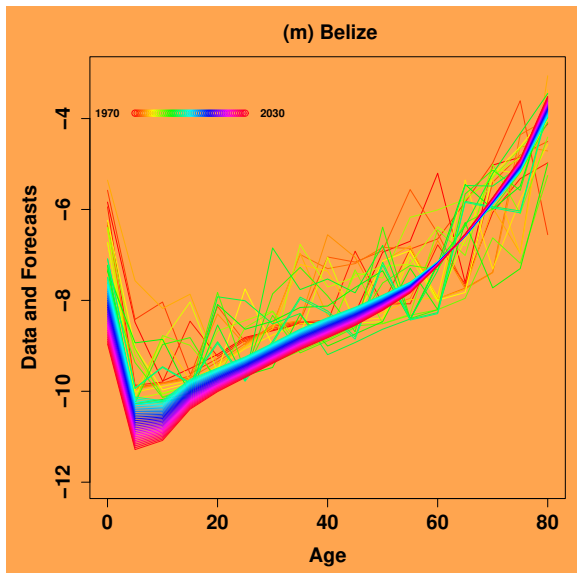
Mortality from Respiratory Infections, males, $\sigma = 0.50$

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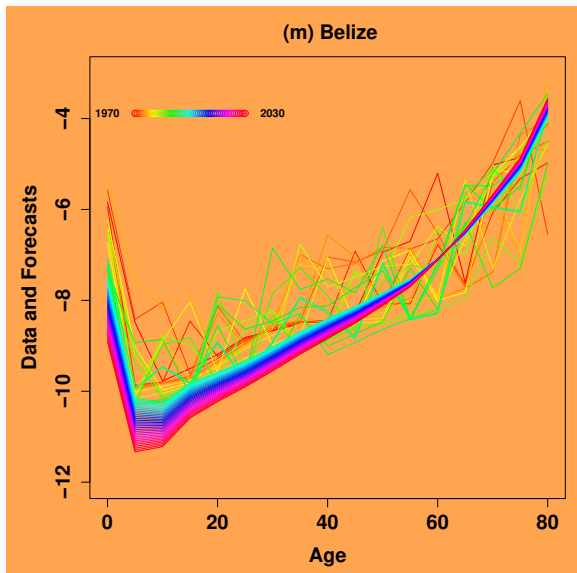
Mortality from Respiratory Infections, males, $\sigma = 0.38$

Smoothing over Age Groups



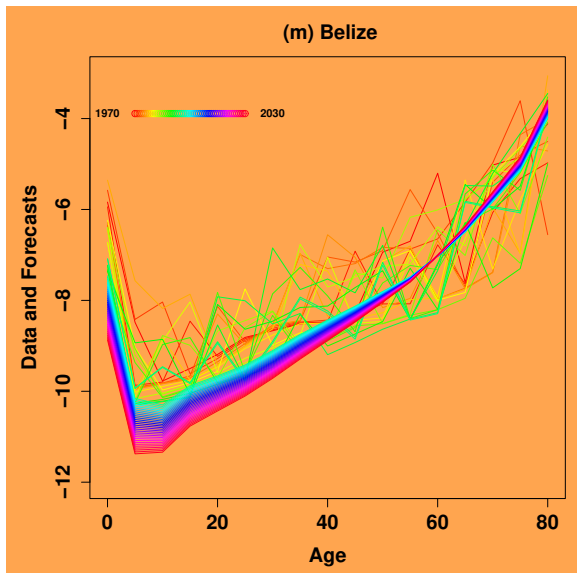
Mortality from Respiratory Infections, males, $\sigma = 0.28$

Smoothing over Age Groups



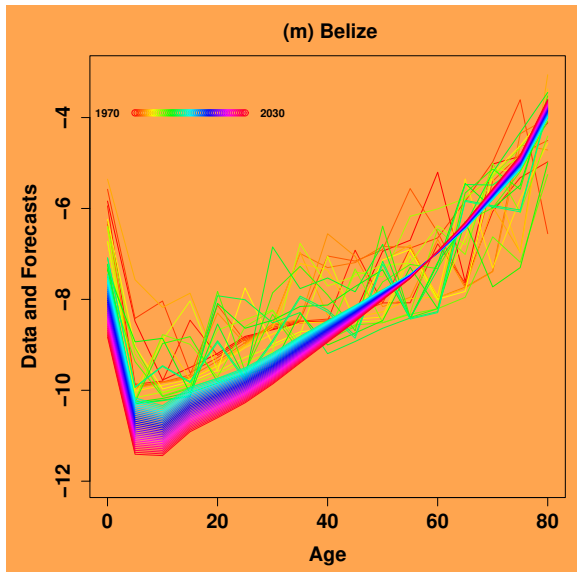
Mortality from Respiratory Infections, males, $\sigma = 0.21$

Smoothing over Age Groups



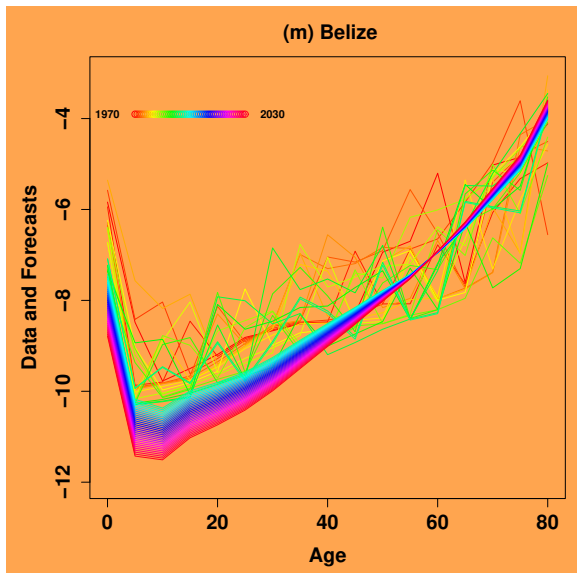
Mortality from Respiratory Infections, males, $\sigma = 0.16$

Smoothing over Age Groups



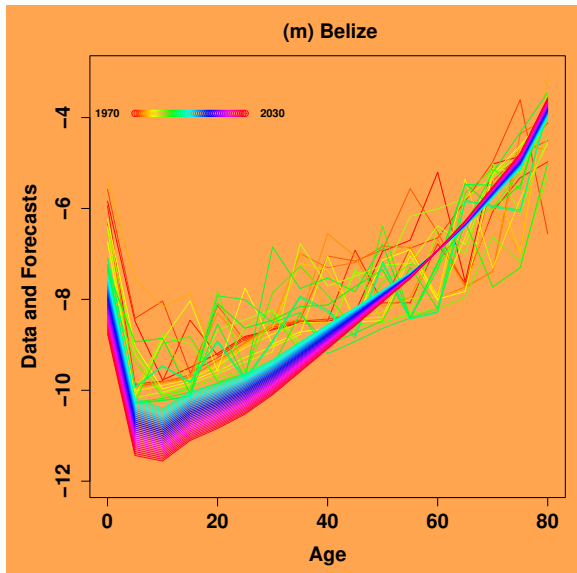
Mortality from Respiratory Infections, males, $\sigma = 0.12$

Smoothing over Age Groups



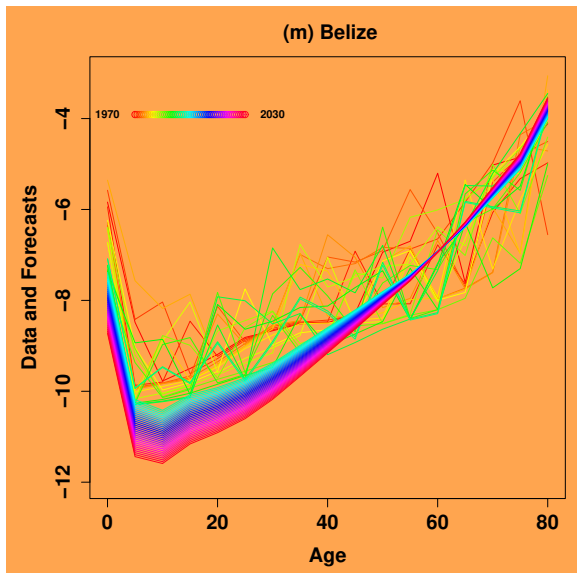
Mortality from Respiratory Infections, males, $\sigma = 0.09$

Smoothing over Age Groups



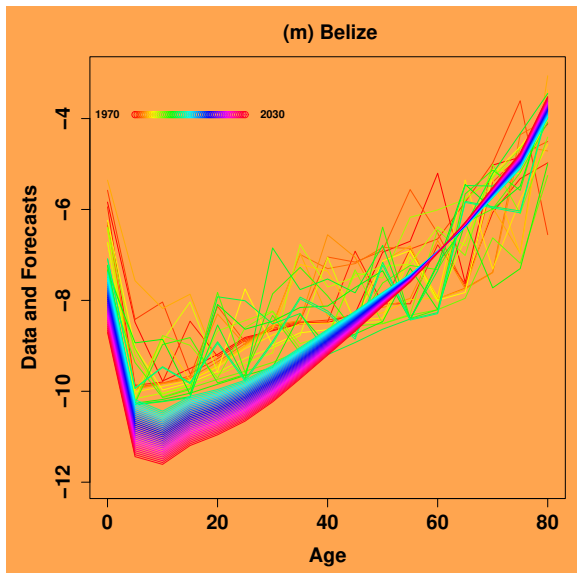
Mortality from Respiratory Infections, males, $\sigma = 0.07$

Smoothing over Age Groups



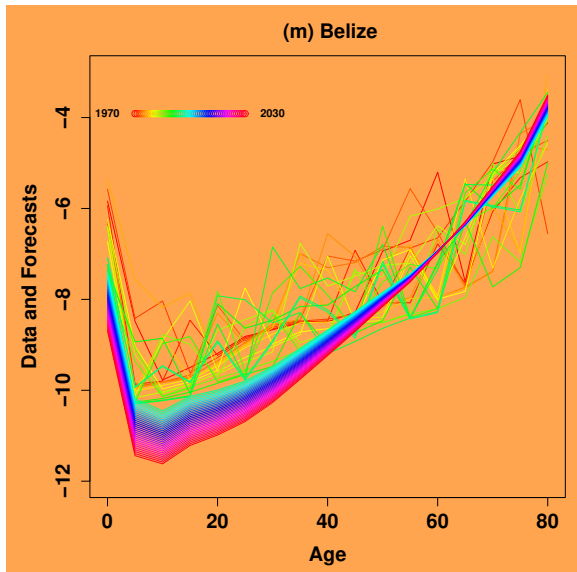
Mortality from Respiratory Infections, males, $\sigma = 0.05$

Smoothing over Age Groups



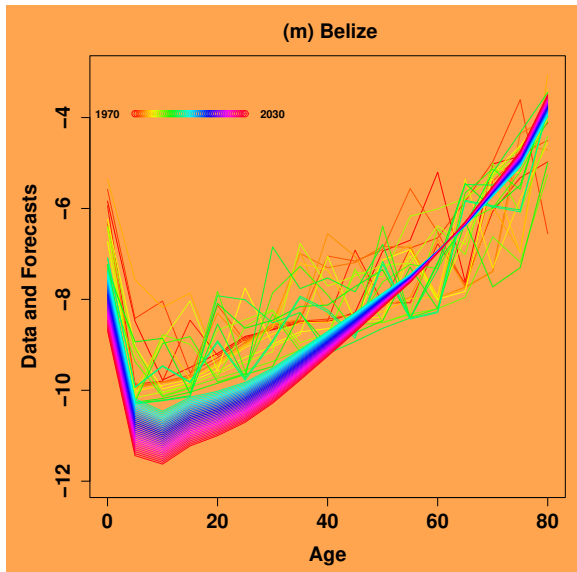
Mortality from Respiratory Infections, males, $\sigma = 0.04$

Smoothing over Age Groups



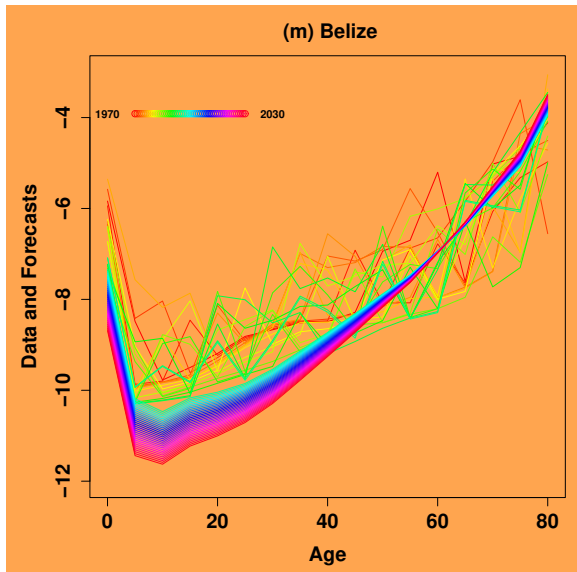
Mortality from Respiratory Infections, males, $\sigma = 0.03$

Smoothing over Age Groups



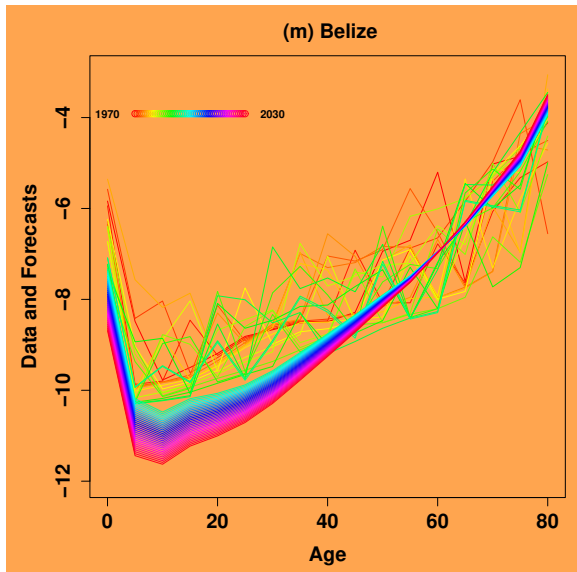
Mortality from Respiratory Infections, males, $\sigma = 0.02$

Smoothing over Age Groups



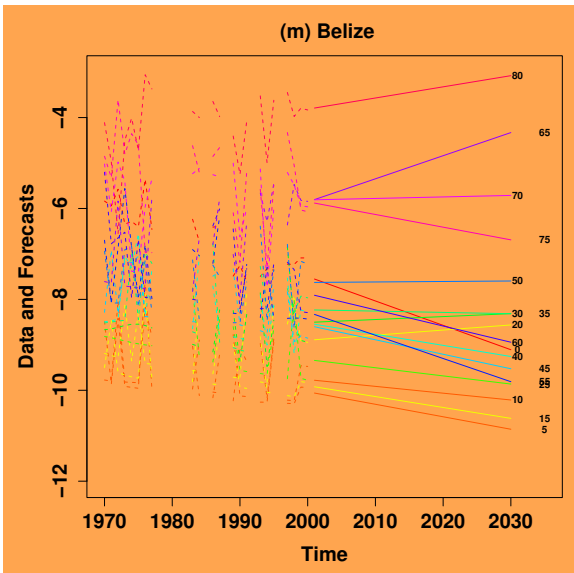
Mortality from Respiratory Infections, males, $\sigma = 0.01$

Smoothing over Age Groups



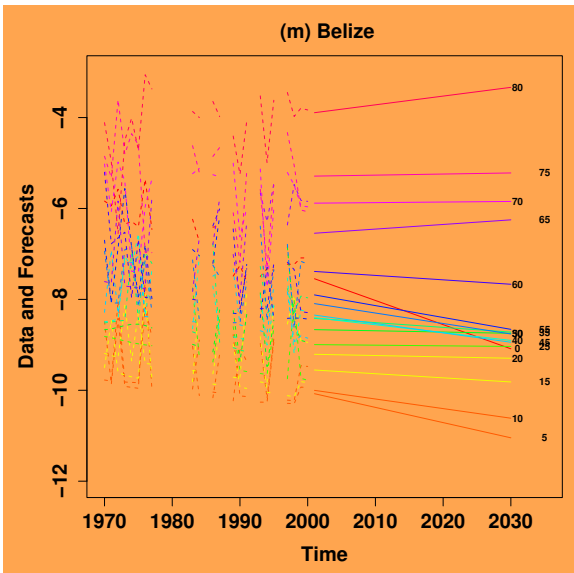
Mortality from Respiratory Infections, males

Least Squares



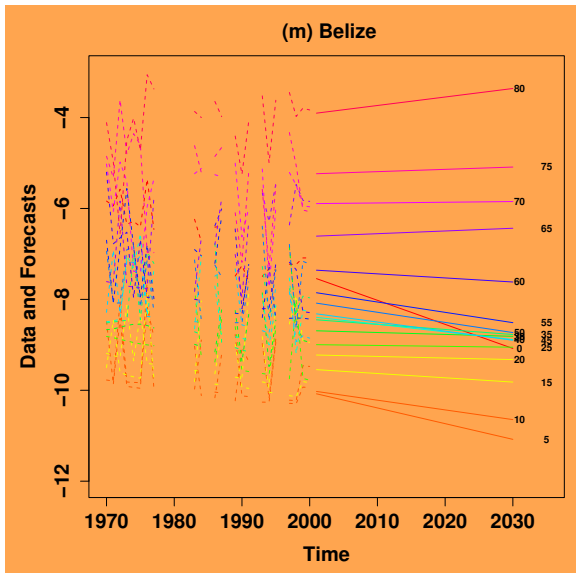
Mortality from Respiratory Infections, males, $\sigma = 2.00$

Smoothing over Age Groups



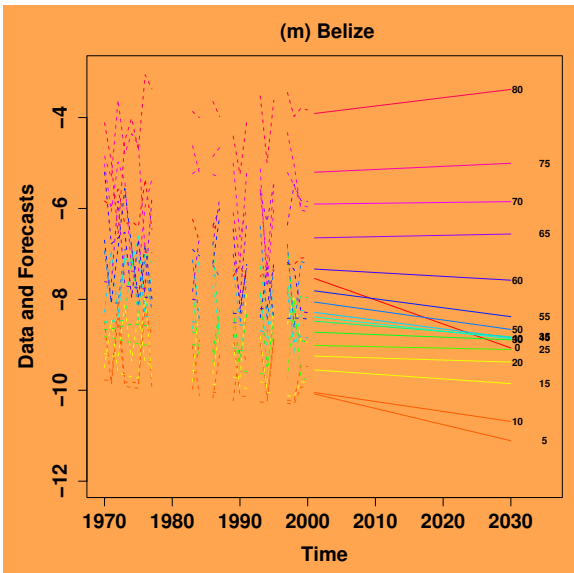
Mortality from Respiratory Infections, males, $\sigma = 1.51$

Smoothing over Age Groups



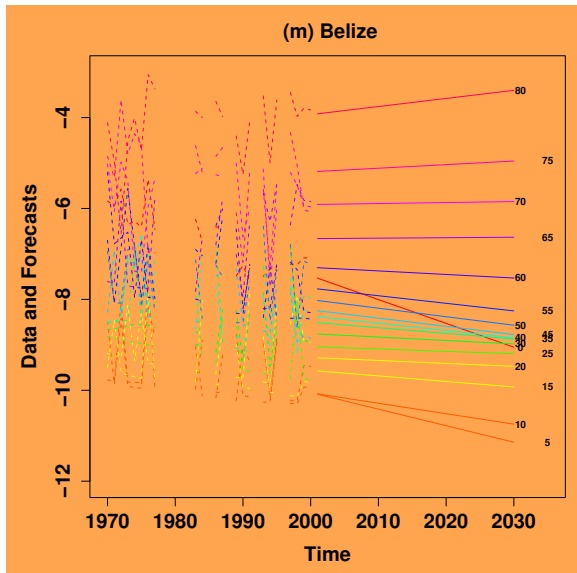
Mortality from Respiratory Infections, males, $\sigma = 1.15$

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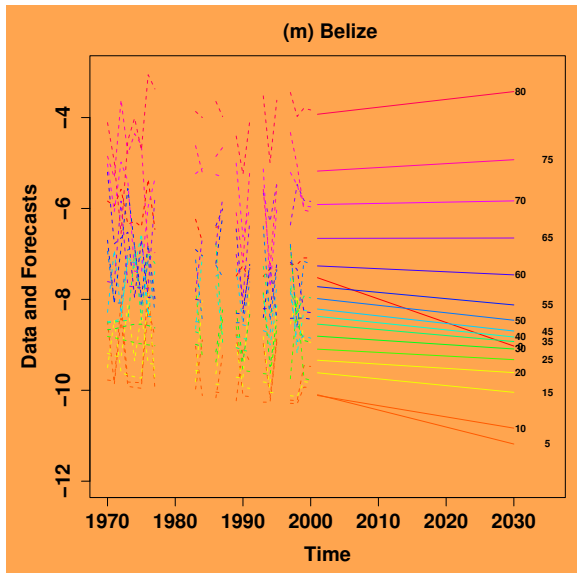
Mortality from Respiratory Infections, males, $\sigma = 0.87$

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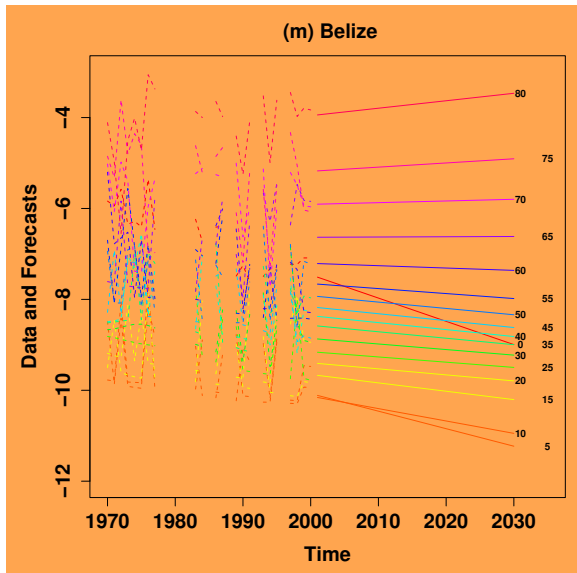
Mortality from Respiratory Infections, males, $\sigma = 0.66$

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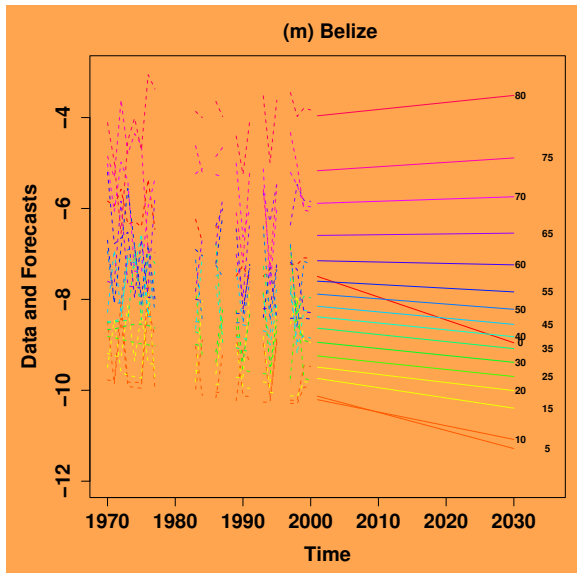
Mortality from Respiratory Infections, males, $\sigma = 0.50$

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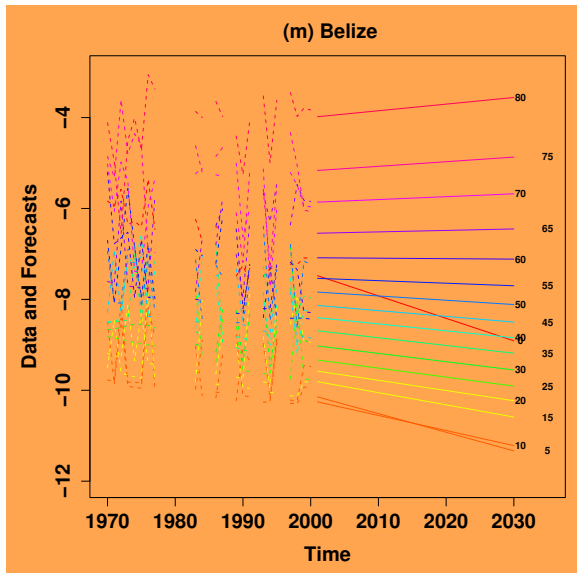
Mortality from Respiratory Infections, males, $\sigma = 0.38$

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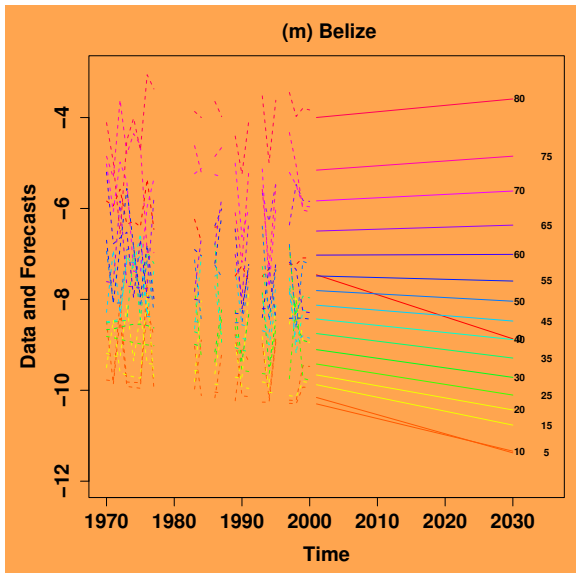
Mortality from Respiratory Infections, males, $\sigma = 0.28$

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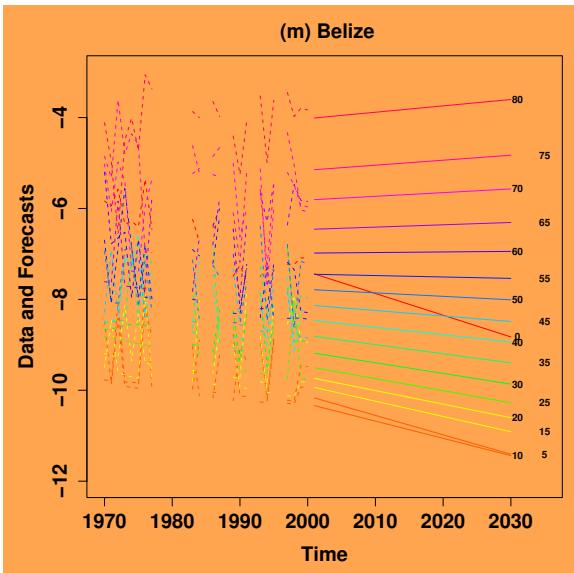
Mortality from Respiratory Infections, males, $\sigma = 0.21$

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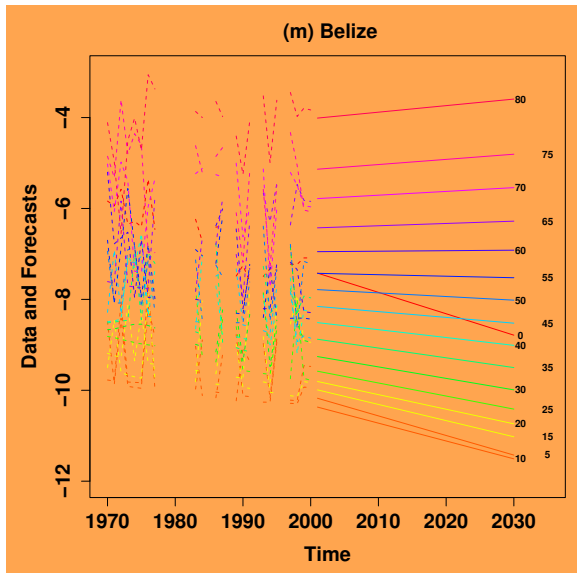
Mortality from Respiratory Infections, males, $\sigma = 0.16$

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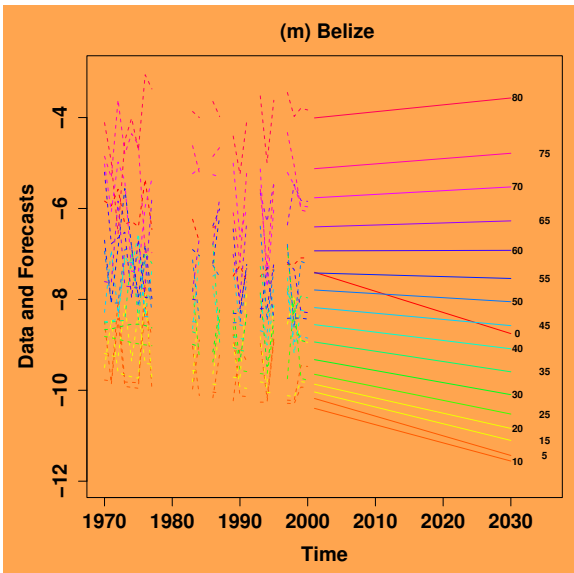
Mortality from Respiratory Infections, males, $\sigma = 0.12$

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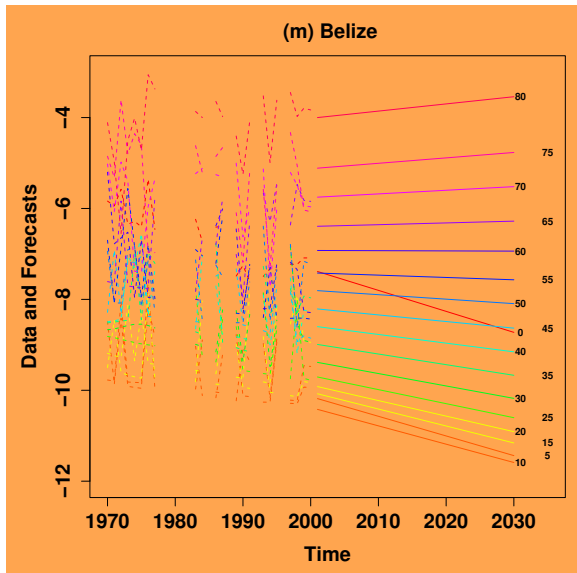
Mortality from Respiratory Infections, males, $\sigma = 0.09$

Smoothing over Age Groups



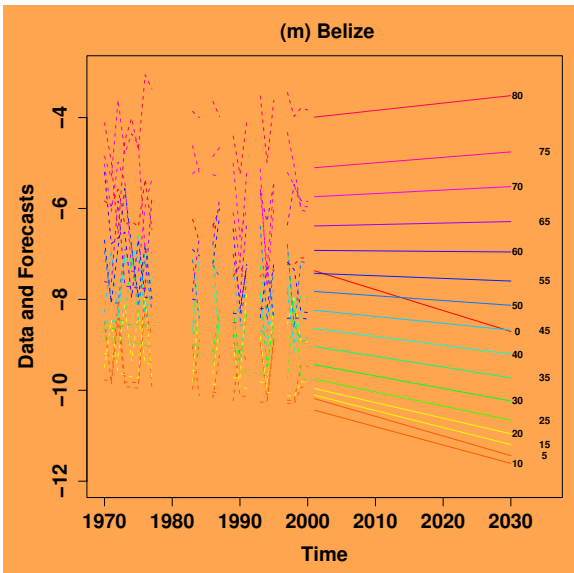
Mortality from Respiratory Infections, males, $\sigma = 0.07$

Smoothing over Age Groups



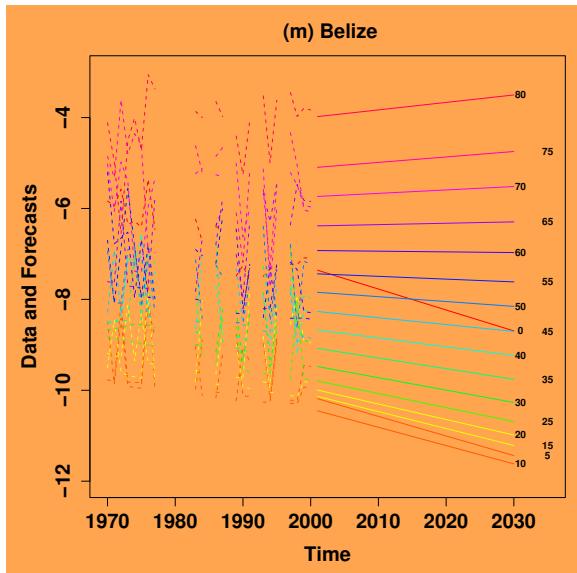
Mortality from Respiratory Infections, males, $\sigma = 0.05$

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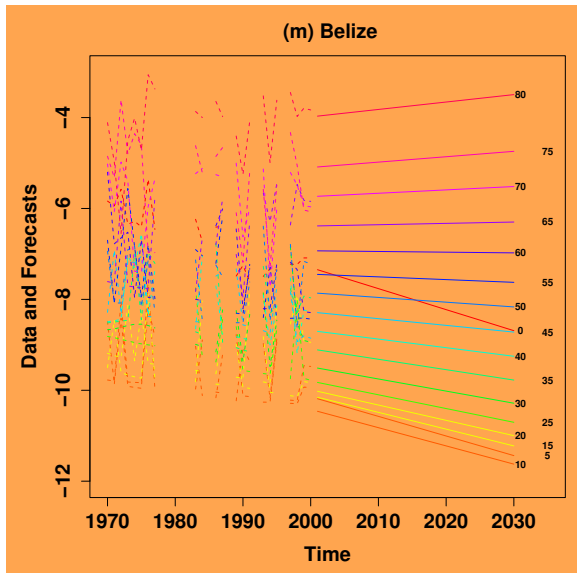
Mortality from Respiratory Infections, males, $\sigma = 0.04$

Smoothing over Age Groups



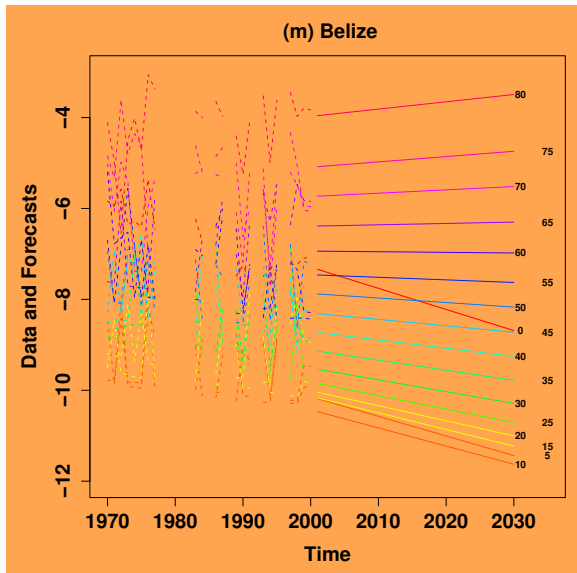
Mortality from Respiratory Infections, males, $\sigma = 0.03$

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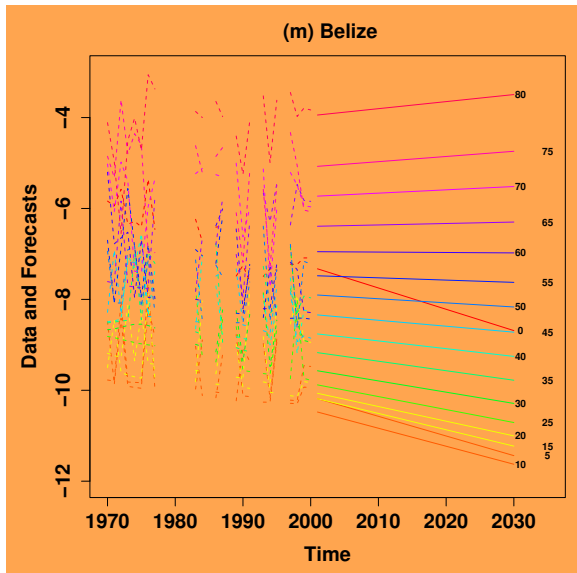
Mortality from Respiratory Infections, males, $\sigma = 0.02$

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Mortality from Respiratory Infections, males, $\sigma = 0.01$

Smoothing over Age Groups



Smoothing Trends over Age Groups

Smoothing Trends over Age Groups

Log-mortality in Belize males from respiratory infections

Smoothing Trends over Age Groups

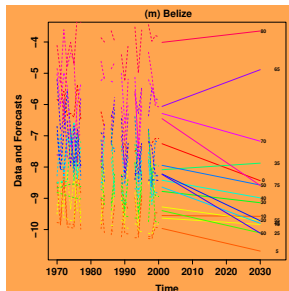
Log-mortality in Belize males from respiratory infections

Least Squares

Smoothing Trends over Age Groups

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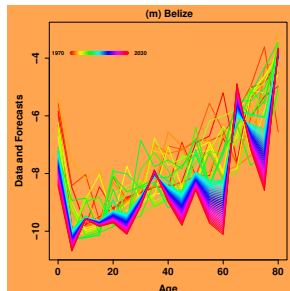
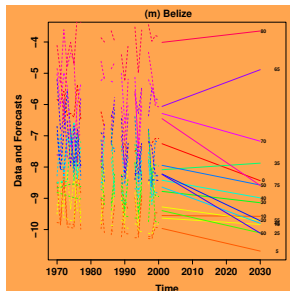
Least Squares



Smoothing Trends over Age Groups

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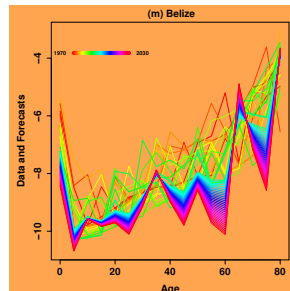
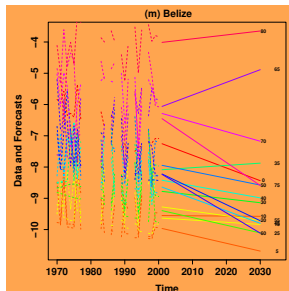
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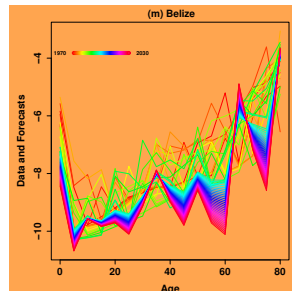
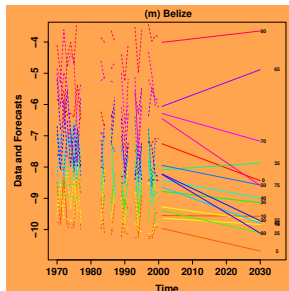


Smoothing
Age Groups

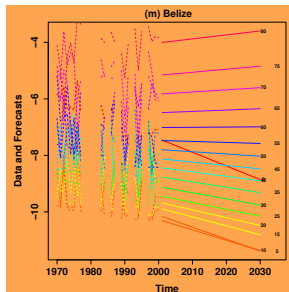
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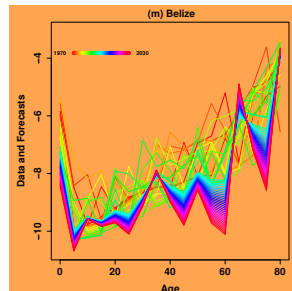
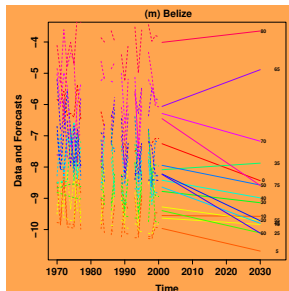
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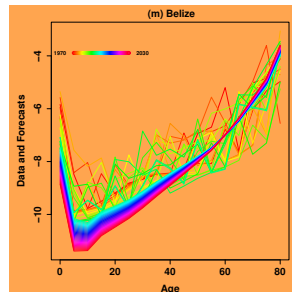
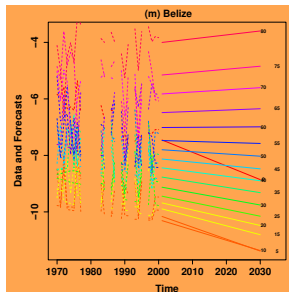
Smoothing Trends over Age Groups

Log-mortality in Belize males from respiratory infections

Least Squares



Smoothing
Age Groups



Smoothing Trends over Age Groups and Time

Smoothing Trends over Age Groups and Time

Log-Mortality in Bulgarian males from respiratory infections

Smoothing Trends over Age Groups and Time

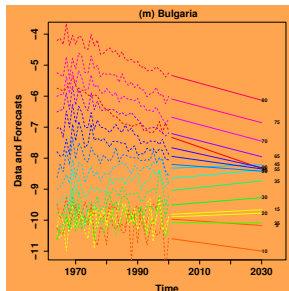
Log-Mortality in Bulgarian males from respiratory infections

Least Squares

Smoothing Trends over Age Groups and Time

Log-Mortality in Bulgarian males from respiratory infections

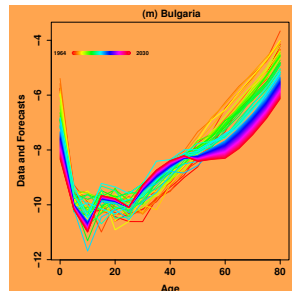
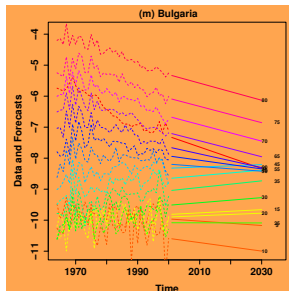
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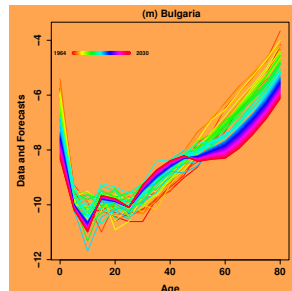
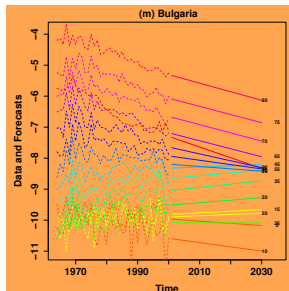
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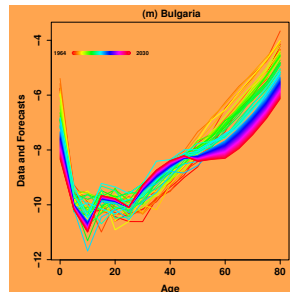
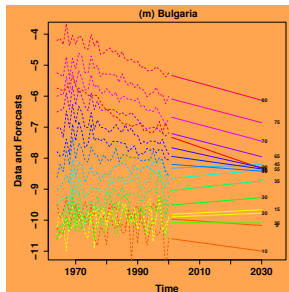


Smoothing
Age and Time

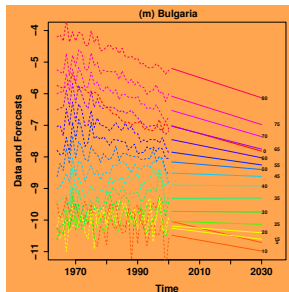
Smoothing Trends over Age Groups and Time

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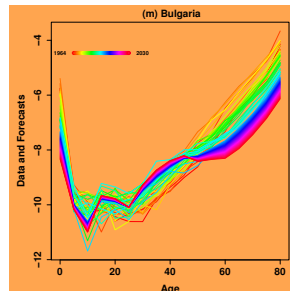
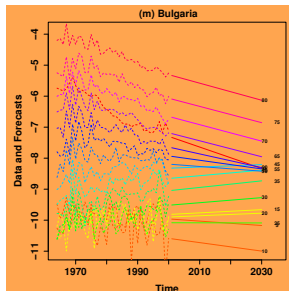
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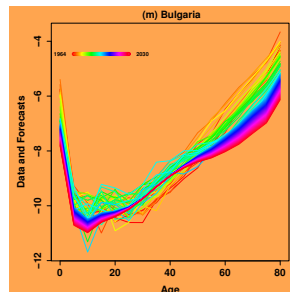
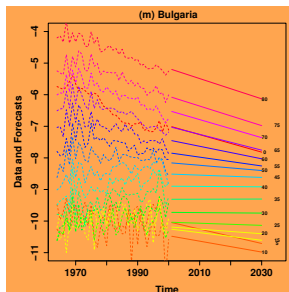
Smoothing Trends over Age Groups and Time

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Smoothing
Age and Time



Using Covariates (GDP, tobacco, trend, log trend)

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Lung cancer in Korean Males

Using Covariates (GDP, tobacco, trend, log trend)

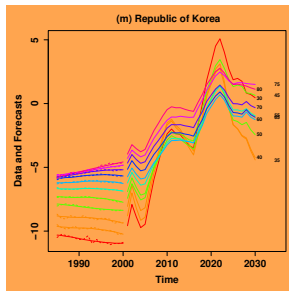
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Using Covariates (GDP, tobacco, trend, log trend)

Lung cancer in Korean Males

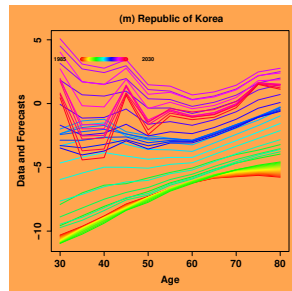
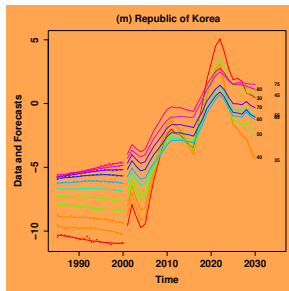
Least Squares



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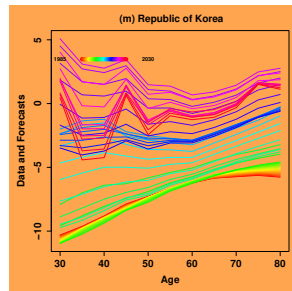
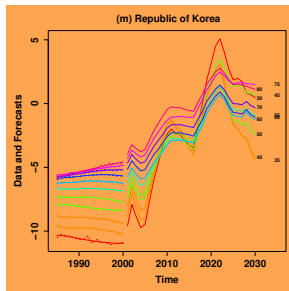
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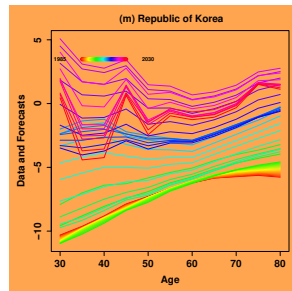
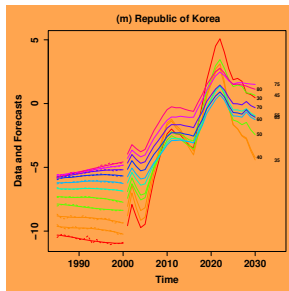


Smooth over age,
time, age/time

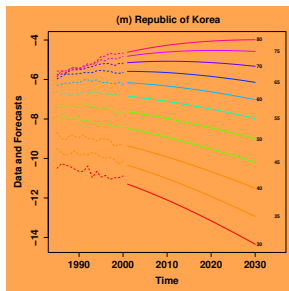
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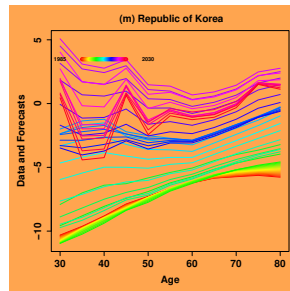
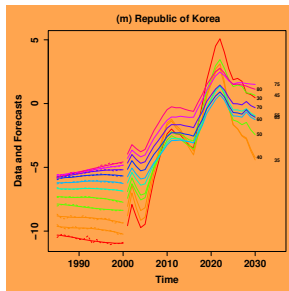
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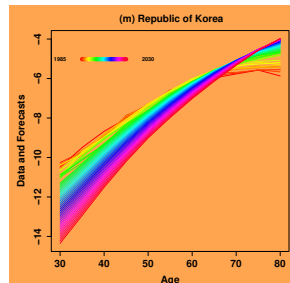
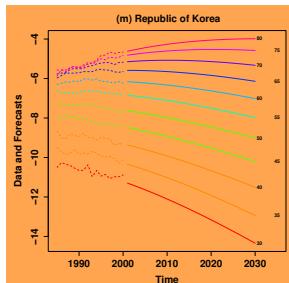
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Using Covariates (GDP, tobacco, trend, log trend)

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Lung cancer in Males, Singapore

Using Covariates (GDP, tobacco, trend, log trend)

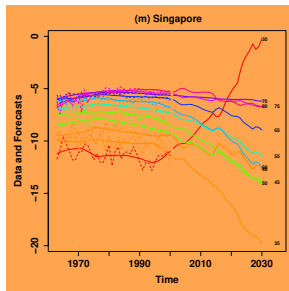
Lung cancer in Males, Singapore

Least Squares

Using Covariates (GDP, tobacco, trend, log trend)

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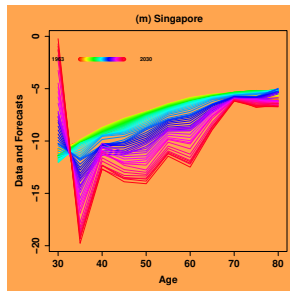
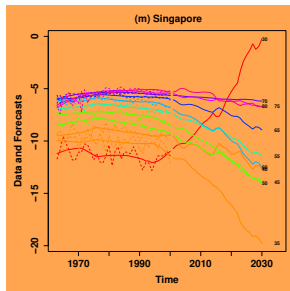
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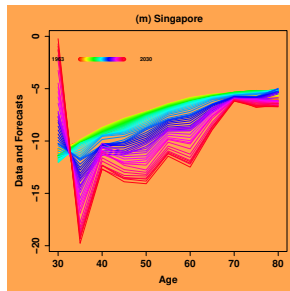
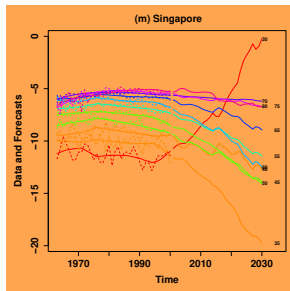
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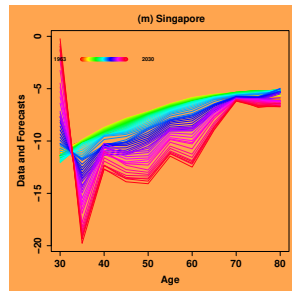
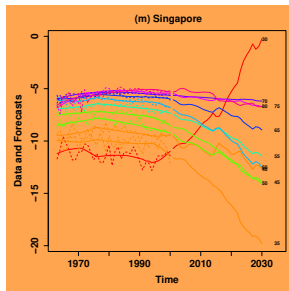


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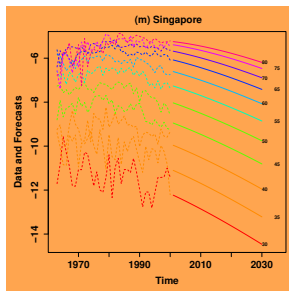
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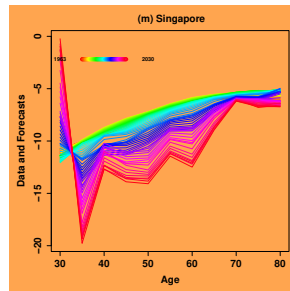
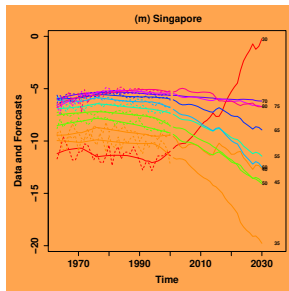
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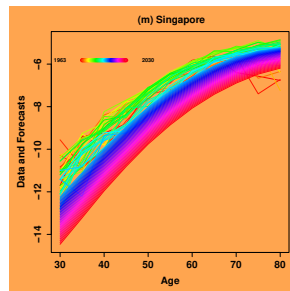
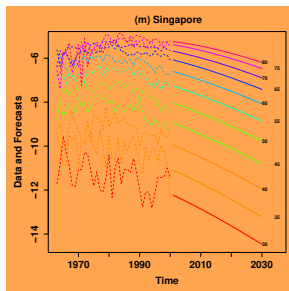
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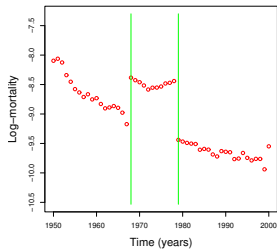


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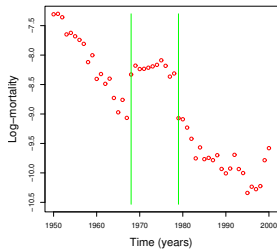


What about ICD Changes?

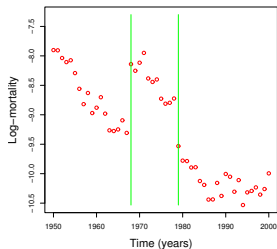
Other Infectious Diseases : USA , age 0 (m)



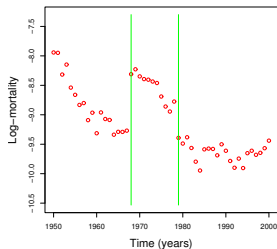
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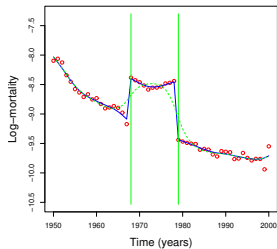


Other Infectious Diseases : United Kingdom , age 0 (m)

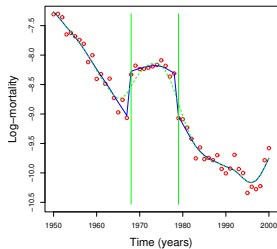


Fixing ICD Changes

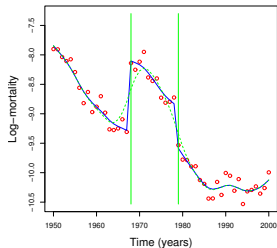
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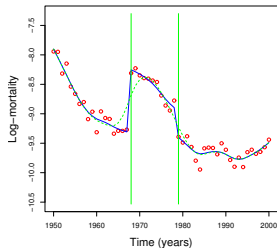
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A book manuscript, YourCast software, etc.

<http://GKing.Harvard.edu>

Preview of Results: Out-of-Sample Evaluation

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Mean Absolute Error in Males (over age and country)

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	<u>% Improvement</u>	
	Over Best Previous	to Best Conceivable
Cardiovascular	22	49
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- Does *considerably* better with **more informative covariates**

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where we have defined:

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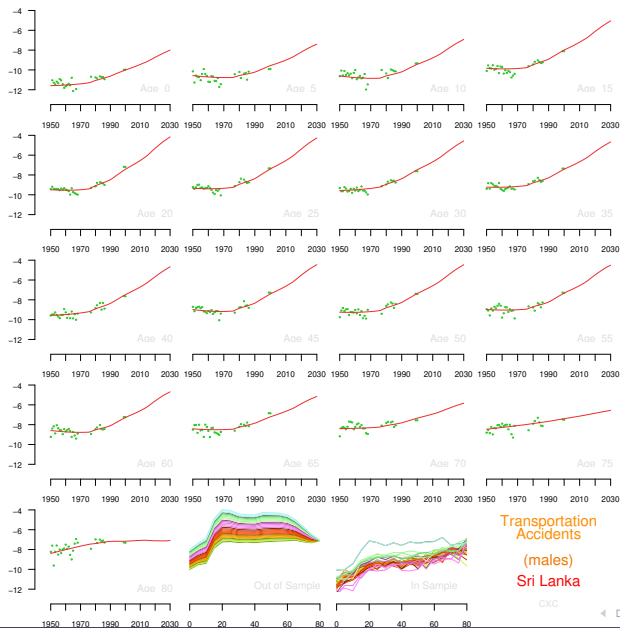
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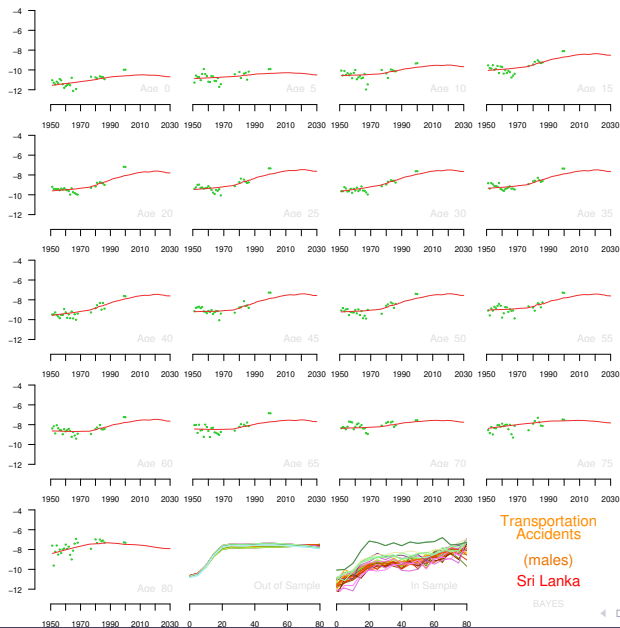
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Without Country Smoothing



With Country Smoothing



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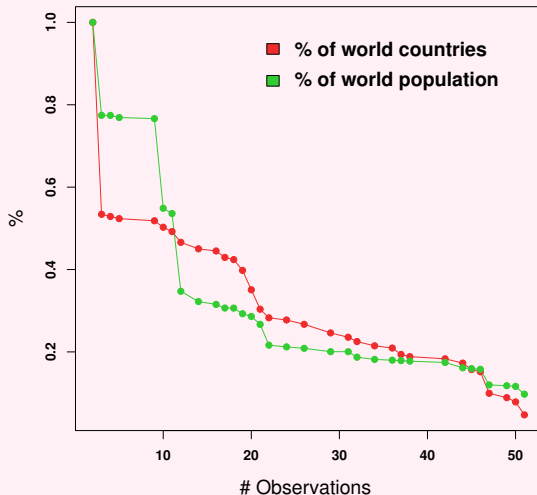
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Many Short Time Series

Coverage of WHO data base (age specific, all causes)



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Lung Cancer	0.36	0.27	0.17	24	47
Transportation	0.37	0.31	0.18	16	31
Respiratory Chronic	0.45	0.39	0.26	13	30
Other Infectious	0.55	0.48	0.32	12	30
Stomach Cancer	0.30	0.27	0.20	8	24
All-Cause	0.17	0.15	0.08	12	22
Suicide	0.31	0.29	0.18	7	17
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Preview of Results: Out-of-Sample Evaluation

Mean Absolute Error in Males (over age and country)

	Mean Absolute Error			% Improvement	
	Best Previous	Our Method	Best Conceivable	Over Best Previous	to Best Conceivable
Cardiovascular	0.34	0.27	0.19	22	49
Lung Cancer	0.36	0.27	0.17	24	47
Transportation	0.37	0.31	0.18	16	31
Respiratory Chronic	0.45	0.39	0.26	13	30
Other Infectious	0.55	0.48	0.32	12	30
Stomach Cancer	0.30	0.27	0.20	8	24
All-Cause	0.17	0.15	0.08	12	22
Suicide	0.31	0.29	0.18	7	17
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- Each row averages 6,800 forecast errors (17 age groups, 40 countries, and 10 out-of-sample years).

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- The new method out-performs with the same covariates, for most countries, causes, sexes, and age groups.
- Does much better with better covariates